Break-Apart Arrays

TEACHER KEY

3rd Grade Math · Interactive Math · Explore & Reflect

Get started

Split an array of unit squares into two smaller arrays to find hard facts from easier ones. Rotate the array to see the commutative property. Open the tool and use it to answer each question.

Explore

1. Break 6 × 7 into 6 × 5 + 6 × 2. Find each part and the total.

30 + 12 = 42

2. Break 8 × 6 into two equal parts. Write the equation.

8 × 3 + 8 × 3 = 24 + 24 = 48

3. Use a 5s fact to find 4 × 9.

4 × 5 + 4 × 4 = 20 + 16 = 36

4. Rotate 7 × 4. What is 4 × 7?

28 — the same as 7 × 4

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Explore more

5. What is the area of a rectangle 3 units by 8 units?

24 square units

6. Fill in the blank: 9 × 6 = 9 × 5 + 9 × __

1

Reflect

7. Reflect: Why does breaking an array apart give the same total?

No squares are added or taken away — the two smaller arrays still cover the whole array.

Teacher notes

- Ask: Which split makes this fact easiest for you? There are several good answers.
- Connect to area (3.MD.C.7c): the same picture is a rectangle split into two rectangles.
- Record students' splits on the board to show that different splits give the same product.

Standards: 3.OA.B.5, 3.OA.C.7, 3.MD.C.7c. Accept any correct answer where several are possible, and ask students to explain with the tool's picture.